KOR STRUCTURAL · DRAFTING STANDARDS

The CAD Detail button — one master, drawn once

For Nelson Yu · from Ian Lalonde · 23 October

You told me converting doesn't work in either direction, so the typical details are drawn twice. This is the thing I built to take the Revit half of that away: a button in Revit that brings a detail in from the CAD master and prints the way KOR.ctb plots it. You keep drawing in CAD exactly as you do now.

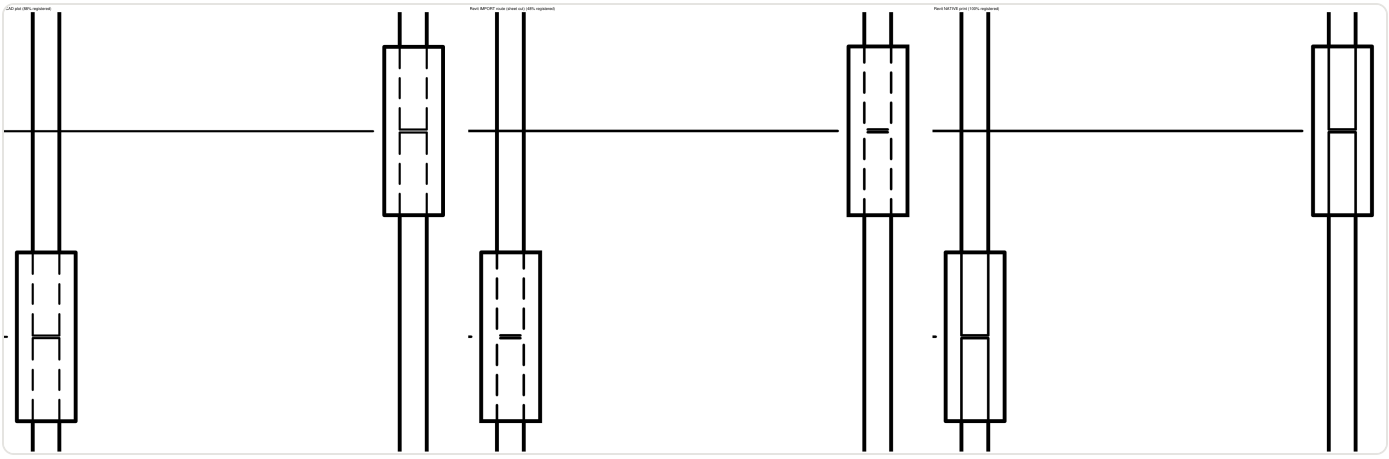
WHAT HAPPENS WHEN YOU PRESS IT

STEP	WHAT	NOTES
1. Open a drafting view	In the job, in whatever Revit year the job is in.	It only ever lands in a drafting view — never on a plan or a sheet.
2. Pick the detail	The file dialog opens on `I:\Kor Structural\002 Typical Details\Typical Details Blocks`. Choose the detail's drawing.	The same blocks you keep today; nothing is copied anywhere else.
3. AutoCAD reads it the way the plotter does	In the background, on your own PC, a copy is prepared: every line moves onto a layer named for its printed pen (`KOR 0.3`, `KOR 0.3 SCREEN 15`), wide polylines become filled bars, hidden and dashed lines keep their dashes, no-plot layers stay unprinted, blocks keep their layer-0 and BYBLOCK meaning.	Needs AutoCAD on the PC. Takes a few seconds.
4. Revit imports the copy	Into your view, at the detail's scale, each pen given its Revit line weight and its black or grey.	No yellow text, no magenta concrete, no flat line weights.
5. Put it on a sheet	Like any drafting view: drag it onto the sheet, number and title as usual.	That is how all 190 were printed for the comparison below.

WHAT IT WAS CHECKED AGAINST — ALL 190 TYPICAL DETAILS

Every detail on the big sheet was brought in (190 of 190, none refused), printed from a Revit sheet, and compared with the CAD plot of the same block under KOR.ctb — not by eye only, by measuring, on 23 October. 181 of the 190 match on everything measured; the other 9 differ in text only, and those are the three questions at the end.

WHAT	RESULT	WHAT IT MEANS
Linework	`190 of 190 details: every line within 1/1000" of CAD's`	Nothing moved, nothing missing, nothing extra.
Dashes	`190 of 190: all 2,098 dashed and hidden lines (in 148 details) dashed at CAD's length`	Until today Revit printed them solid — the last fault found, fixed 23 October. Where a dash begins along a line is Revit's choice, so the dashes do not always start at the same point as CAD's; same line to the eye.
Bars and fills	`filled bars and screened concrete on both prints`	The 13% magenta concrete prints as the same light grey; rebar dots are dots, not rings.
Text	`181 of 190 details word-for-word; 9 differ`	The differences are the font (CAD's SHX letters become a Revit font), the Ø symbol, and word spacing — see "Need from you".



S302 detail 11, coupler sleeves, one window at 10 px per point. Left: the CAD plot. Middle: Revit, via the button. Right: a Revit-native rebuild (an experiment, not the button) — its hidden lines are solid, which is why it is not the one going to you.

WHAT THIS MEANS FOR THE TYPICAL-DETAILS CATALOGUE

The CAD master stays the master. The catalogue in the KOR app lists every typical detail with its CAD block, who approved it (Jim) and when; the Revit copy is no longer something anyone draws — the button makes it from the CAD block, in the job's own Revit year, when a job needs it. So: one change, in CAD, approved once, and the Revit side follows. The Typical Details Container in R23 and the details merged into the Concrete Tower template stop being things to keep up to date. Cover sheets are a different matter — engineers change those per job — and nothing here touches them.

WHAT REVIT CANNOT DO, SO YOU KNOW BEFORE YOU LOOK

LIMIT	WHERE YOU'D SEE IT	SIZE
Font sizes round to ½ pt	Revit's PDF print rounds text height; 2.5 mm text prints at the nearest half point.	Under 0.1 pt on lettering; invisible on paper, measurable with a ruler on the file.
SHX fonts	CAD's stroke fonts (simplex, romans) are drawn in a Revit TrueType font instead.	The letters are the same words at the same size; the strokes look slightly different.
Splines	A curve drawn as a spline comes in as short straight pieces.	Within 1/1000"; you cannot see it at any plotted scale.
Linetypes with a symbol in them	The W.W.M. linetype draws an X along the line in CAD; Revit's line patterns carry dashes only, so the X marks are not there.	One line in one detail of the 190 (D504-20, built-up pool deck).

NEED FROM YOU

1. **Sit with Ian at KOR-302N for twenty minutes.** Three details printed both ways on the desk: D301-01 (basement wall — grey concrete, hidden bars), S302-11 (the coupler), one wood-frame detail. Say what is wrong.
2. **Pick the Revit font** for CAD's SHX text. Whatever you choose is applied to every detail the button brings in.
3. **Confirm the pen table.** KOR.ctb's widths are mapped onto Revit's line weights 1–16; Ian brings the mapping on one page.

THEN

1. The button goes onto your PC (KOR-307-N), alongside the tools you already have. Nothing is removed.
2. You try it on a live job in its own Revit year and tell Ian what you hit.
3. The catalogue's approve step with Jim is switched on for CAD changes, so a detail changed in CAD is the one everybody gets.